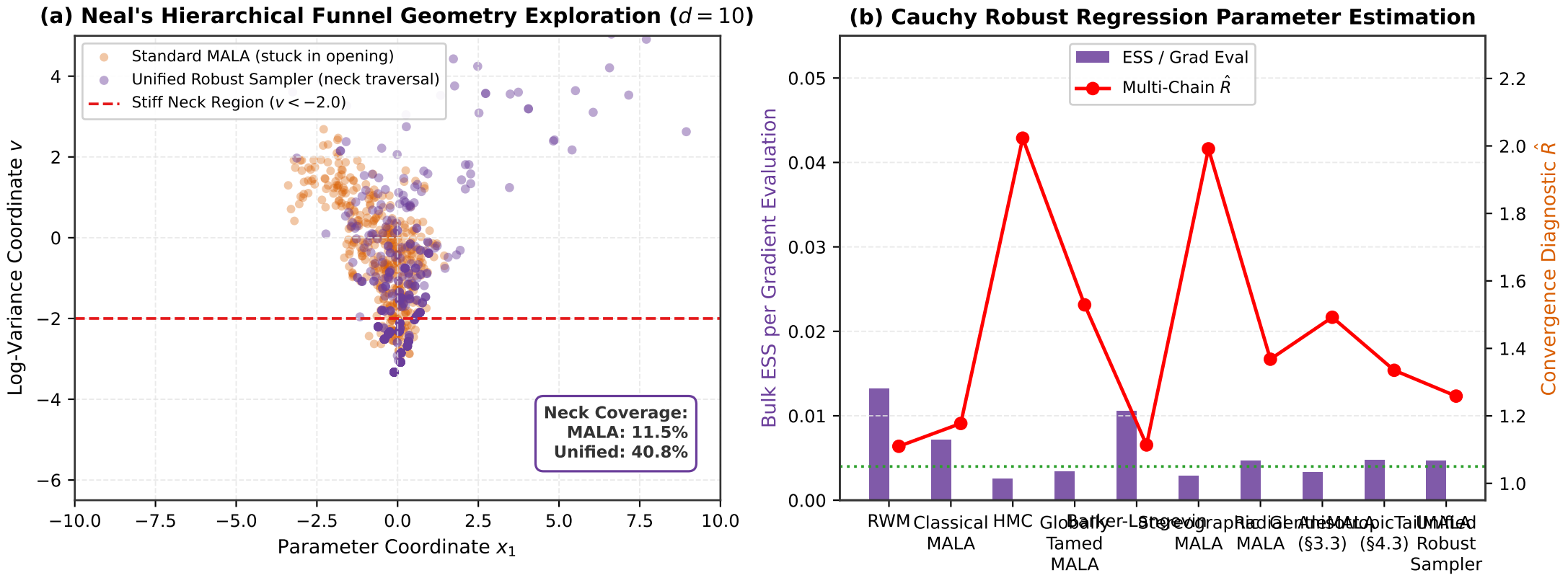
(a) Neal's Hierarchical Funnel Geometry Exploration ($d = 10$)
Standard MALA (stuck in opening)
Unified Robust Sampler (neck traversal)
Stiff Neck Region ($v < -2.0$)
Neck Coverage:
MALA: 11.5%
Unified: 40.8%
Log-Variance Coordinate v
Parameter Coordinate x_1
(b) Cauchy Robust Regression Parameter Estimation
ESS / Grad Eval
Multi-Chain $\hat{R}$
Bulk ESS per Gradient Evaluation
Convergence Diagnostic $\hat{R}$
RWM
Classical MALA
HMC
Globally Tamed MALA
Barker-Langevin
Stereographic
Riemannian MALA
Anisotropic (§3.3)
Anisotropic (§4.3)
Tail-Tuned Robust
Unified